

Syllabus

MATH 241 (01, 02): Probability, Fall 2026

In this webpage, you will find the syllabus for this course (01, 02). A .pdf version of this syllabus can be accessed [here](#).

Overview

Contact

- Instructor: Jericho Lawson
- Email: jblawson@vassar.edu
- Office Hours:
 - Mondays, 1-2:30pm
 - Tuesdays, 3-4:30pm
 - Rocky 106A

Location/Time

- 01: Tuesdays & Thursdays, 12 - 1:15pm in Rockefeller Hall 203
- 02: Tuesdays & Thursdays, 1:30 - 2:45pm in Rockefeller Hall 210

Note: You MUST attend the section you are signed up for.

Description

A first course in mathematical probability that covers topics including combinatorics, discrete and continuous random variables, distribution functions, joint distributions, independence, properties of expectations, and basic limit theorems. No prior background in statistics or probability is expected.

Course Prerequisite: MTH 126-127 (i.e. Calculus II), or equivalent.

Learning Outcomes

By the end of this course, students will be able to do the following:

- *Conceptualize* various foundational ideas in probability theory, including the axioms of probability, combinatorial probability, conditional probability, and independence.
- *Work* with discrete and continuous probability spaces through the use of random variables, with their associated distribution functions and moments, recognizing patterns and similarities between different families of probability distributions.
- *Apply* tail inequalities, the law of large numbers, and the central limit theorem to study and approximate distribution properties.
- *Use* technical tools appropriately, such as R, RStudio, and generative AI, to simulate concepts and problems in probability theory, as well as perform meaningful calculations.
- *Communicate* statistical results and real-world impacts to various stakeholders by written, visual, and verbal means.

Textbooks

- Blitzstein, J. K., & Hwang, J. (2019). *Introduction to Probability (2nd ed.)*. Chapman & Hall/CRC. <http://probabilitybook.net>
 - This textbook is **free** to use online!
 - More resources can be found through this [course website](#), for which our textbook is based on.

Technology

- For coding: **R** and **RStudio**
 - [Instructions](#)
- Assignment turn-ins: **Gradescope**
 - Entry code for 01: KNE4DG
 - Entry code for 02: 7KRWXN
- Course management: **Moodle**
 - 01 [course](#)
 - 02 [course](#)
 - Will be used seldomly.
- Course materials: [Course Website](#)
- Discussions and announcements: **Slack**

- 01 [course](#)
- 02 [course](#)

Grading and Expectations

Breakdown

Formative		Summative	
Engagement	5%	Checkpoints (~6)	20%
Homework (~10)	20%	Project	15%
		Exams (3)	40%

A standard grading system will be used here. Cumulative numerical averages of 90-100 are guaranteed at least an A-, 80-89 at least a B-, and 70-79 at least a C-.

Engagement (5%)

Success in this course will involve strong engagement with course material, both inside and outside the classroom. This includes the assignment of readings and materials, interacting with content and peers in the classroom, and bringing concerns and questions to the instructor when needed.

Readings and materials will be assigned outside of class for you to prepare for the following class. This will involve chapters from the textbook, external readings, videos, etc. We will discuss these materials in-class, so be prepared to discuss what you learned and saw.

Attendance is strongly expected and will be incorporated into your engagement grade. Additionally, when you are in class, you are expected to be actively engaged. Below is a rubric for how engagement will be scored:

5

Student consistently shows up for classes in their entirety, with no more than three classes missed. When in class, the student participates regularly with other students and the instructor, having prepared for class by reading materials outside of class.

4

Student consistently shows up for classes, with no more than three classes missed. The student mostly participates with other students and the instructor, with some preparedness for class by reading materials outside of class.

3

Student shows up for classes relatively often. The student somewhat participates with other students and the instructor, with occasional preparedness for class by reading materials outside of class.

2

Student misses classes more than on occasion, with some notice to the instructor. The student occasionally participates with other students and the instructor, with at least minimal preparedness for class by reading materials outside of class.

1

Student consistently misses classes, without any notice to the instructor. The student might participate with other students and the instructor, with at least minimal preparedness for class by reading materials outside of class.

0

Student consistently misses classes, without any notice to the instructor. The student does not participate with other students and the instructor, with no preparedness for class by reading materials outside of class.

You are strongly recommended to bring your laptop to all lectures. We will be using tools such as R and RStudio to work through examples and submit assignments during class.

Checkpoints (20% combined)

Some Friday classes will be dedicated to checkpoints, which are designed to track your progress with concepts, definitions, and applications in this course. Either by yourself or in a group, 2-3 problems will be solved using your notes and R. They will be completed in-class and turned in by the end of the period. Six checkpoints will be administered throughout the semester.

- Checkpoints will be graded based on **correctness**.
- Your *lowest checkpoint* will be dropped from your final grade.

Homework (20% total)

Homework will be assigned on almost a weekly basis, with these problem sets generally due on Fridays by 11:59pm local time. Problem sets consists of calculations, short answers, and inference, which will demonstrate your conceptual understanding and allow you to practice the methodological components. Homework will be submitted as PDFs on Gradescope and can be completed through a typesetting program (e.g. LaTeX) or by-hand.

- Homework will be graded based on **correctness**.
- Your *lowest homework assignment* will be dropped from your final grade.

Project (15%)

A semester-long project will be assigned sometime around weeks 3-5 and due by the end of the semester, with checkpoints assigned along the way. The project is designed to test your ability to successfully complete course objectives as described earlier.

More details to come in later weeks.

Exams (40% total)

Three in-person exams will be administered throughout the course of the semester, with the following dates:

- Friday, 9/29
- Friday, 11/10
- Finals week, TBD

Anticipate these exams to take up the whole allotted time, with short-form questions occupying the whole exam. Your two highest-scored exams are worth 15% each, while your lowest-scored exam will be worth 10%.

Policies

Preparation and Attendance

Expect to spend an average of 10 hours per week on this course, with 2.5 hours spent in-class and 7.5 hours outside of class. Outside of class, you are expected to read through lecture notes, readings, and examples and work on problems (via homework, lab assignments, studying, practice) in preparation for class and beyond. In-class, you are expected to take part in class discussions, exercises, activities, and lectures.

As mentioned earlier, attendance is expected and will be incorporated into your engagement grade. If you are unable to attend class resulting from family/personal difficulties, sickness, religious observations, or school/career-related activities, please let me know as soon as possible.

Do make every effort to arrive to class on time. If you happen/plan to arrive more than 10 minutes late, please inform me ahead of time. Any unexplained tardiness will result in a marked absence. If you must miss a class entirely, you should contact a peer to discuss what was missed.

Collaboration and Class Norms

Inside the classroom, remember that this is a learning space for everyone, and as such, keep these ideals in mind:

- Respect others and yourself as you participate within the class environment.
- Embrace challenge gracefully. Ask questions as they come up.
- Participate to the best of your ability.

Collaboration will occur frequently within the classroom, and naturally outside of class. You are welcome to study and work together as you complete assignments and prepare for exams. However, any written work or code should be your own product (unless otherwise stated). If any code or work is based on those seen from another source (such as StackExchange), please cite the source.

Note: Generative AI is not recommended for doing any part of any assignment (unless otherwise stated), and will not be allowed in any capacity on in-class assignments and exams. See the Generative AI policy for more details.

Extensions and Late Work

You will be granted **3 free late days** to use for homework assignments throughout the semester, with a maximum of one day used on each assignment. Beyond this, late assignments will not be accepted without an accommodation from a class dean.

Academic Honesty

As a student at Vassar College, the college expects all students to be honest and committed to the principles of academic and intellectual integrity in preparation and submission of all course work and examinations. Any cases of academic dishonesty or plagiarism will be reported and may result in a score of 0 for ALL involved persons on the assignment in question. Examples of these behaviors include:

- Submitting work completed by another student as your own.
- Copying and pasting words from sources without quoting and citing the author.
- Paraphrasing material from another source without citing the author.
- Failing to cite your sources correctly.
- Falsifying or misrepresenting information in submitted work.
- Paying another student or service to complete assignments for you.
- Submitting work generated by artificially intelligent tools such as ChatGPT without permission or instruction to do so.

You are encouraged to discuss course material, including assignments, with your classmates. All work you turn in, however, must be your own. This includes both writing and code.

Generative AI

Generative AI is an incredible tool that can be used to facilitate tasks both small and large; however, it should not be a substitute for your own learning of material and concepts in this course. Imagine a generative AI model as another person: if that model is doing something that wouldn't be permissible if they were a human, it probably will not fly here.

Formalizing the analogy, you are open to use generative AI on the following items:

- Formatting/proofreading of outside assignments (excluding the project)
- Creation of study aids for preparation purposes
- Generation of practice problems aside from the assignments

On the flip side, any use of generative AI (unless otherwise stated) is **prohibited** on all in-class assignments and exams. The policy is meant to encourage use for your own learning outside the classroom, while adhering to your own cognitive processes within the classroom for improved learning and comprehension.

Prohibited forms of generative AI usage include but are not limited to:

- Drafting responses to written prompts on any assignment.
- Composing code and comments.
- Answering lab, quiz, or exam questions.
- Conducting analysis of any plots, methods, and results for any assignment.

Any unauthorized use of generative artificial intelligence in this course will be considered a case of academic dishonesty/plagiarism and will be reported.

Community & Support

Code of Conduct

As the instructor for this course, I am committed to making participation in this course a harassment-free experience for everyone, regardless of level of experience, gender, gender identity and expression, sexual orientation, disability, personal appearance, body size, race, ethnicity, age, or religion. Examples of unacceptable behavior by participants in this course include the use of sexual language or imagery, derogatory comments or personal attacks, trolling, public or private harassment, insults, or other unprofessional conduct.

As the instructor, I have the right and responsibility to point out and stop behavior that is not aligned to this. Participants who do not follow the Code of Conduct may be reprimanded for such behavior. Instances of abusive, harassing, or otherwise unacceptable behavior may be reported by contacting the instructor.

All students and the instructor are expected to adhere to this Code of Conduct in all settings for this course: seminars, office hours, and over Slack. This Code of Conduct is adapted from the Contributor Covenant, version 1.0.0, available [here](#).

Q-Center Peer Support Services

This course fulfills the quantitative analysis (QA) requirement for graduation. All Vassar students have access to free, drop-in, peer-to-peer quantitative tutoring at the Quantitative Reasoning Center (Q-Center). Quantitative tutors (Q-Tutors) excel in a variety of STEM courses. They are typically available Sunday-Thursday 3pm-11pm while classes are in session. Q-Tutors who specialize in Mathematics and Physics are located in the Main Library, Room 122 behind the Writing Center. Q-Tutors who specialize in Chemistry and Economics are located in the Main Library, Room 88 near Special Collections. If you have a quantitative question beyond these four disciplines, Q-Tutors are available to attempt to help you with this question or will help direct you to someone else who may be better able to help. Schedules and other important information can be found at <https://offices.vassar.edu/ltrc/qrc/>.

Accommodations

It is my goal for everyone to succeed in this course. If you have personal circumstances that may impact your experience of our classroom, I encourage you to contact the [Office for Accessibility and Educational Opportunity \(AEO\)](#). Students in need of disability (ADA/504) accommodations should schedule an appointment with me early in the semester to discuss any

accommodations for this course that have been approved by the Office for Accessibility and Educational Opportunity, as indicated in your AEO accommodation letter.

Title IX

Vassar College is committed to providing a safe learning environment for all students that is free of all forms of discrimination and sexual harassment, including sexual assault, relationship abuse, and stalking. If you (or someone you know) has experienced or experiences any of these incidents, know that you are not alone. Vassar College has staff members trained to support you in navigating campus life, accessing health and counseling services, providing academic and housing accommodations, helping with legal protective orders, and more.

Please be aware all Vassar faculty members are “responsible employees,” which means that if you tell me about a situation involving sexual harassment, sexual assault, relationship abuse, or stalking, I must share that information with the Title IX Coordinator.

Although I have to make that notification, the Title IX office will only provide outreach by email. You will control how your case will be handled—you don’t have to read or respond to the email, and it is completely up to you whether to pursue a formal complaint. Our goal is to make sure you are aware of the range of options available to you and have access to the resources you need. If you wish to speak to someone privately, you can contact any of the following on-campus resources:

- Counseling Service (<http://counselingservice.vassar.edu>, 845-437-5700)
- Health Service (<http://healthservice.vassar.edu>, 845-437-5800)
- Rachel Gellert, SAVP (Support, Advocacy, & Violence Prevention) director (<http://savp.vassar.edu>, 845-437-7863)
- SAVP (formerly SART) advocate, available 24/7 by calling the CRC at 845-437-7333 and asking for an SAVP advocate.

The SAVP website (<http://savp.vassar.edu>) and the Title IX section of the EOAA website (<https://eoaa.vassar.edu/title-ix/>) have more information, as well as links to both on and off campus resources.

Course Outline

Note that the assignments and topics covered are tentatively scheduled and may be altered as the semester progresses. For a more detailed (and up-to-date) version of what is to come, go to the [Schedule](#) page to see a Google Calendar of all events and assignments.

Week	Dates	Topics	Ch.	Assignments / Notes
1	8/31-9/4	Sets; Sample Spaces	A1, 1.1-1.2	
2	9/7-9/11	Probability and Counting	1.3-1.8	CP #1, Thurs. 9/10
3	9/14-9/18	Conditional Probability; Bayes' Rule	2.1-2.4	CP #2, Thurs. 9/17
4	9/21-9/25	Independence; Review	2.5-2.7	
5	9/28-10/2	Exam #1; Random Variables	3.1, 3.2, 3.6	Exam #1, Tues. 9/29
6	10/5-10/9	Discrete RVs	3.3-3.8	CP #3, Thurs. 10/8
7	10/12-10/16	Expectation; Variance	4.1-4.7	CP #4, Thurs. 10/15
8	10/19-10/23	Fall Break!		<i>No class all week long</i>
9	10/26-10/30	Continuous RVs	5.1-5.7	CP #4, Thurs. 10/29
10	11/2-11/6	Moments; Review	6.1-6.4	
11	11/9-11/13	Exam #2; Joint Distributions	7.1	Exam #2, Tues. 11/10
12	11/16-11/20	Joint Distributions	7.1-7.3	CP #6, Thurs. 11/19
13	11/23-11/27	Conditional Expectation	9.1-9.2	
14	11/30-12/4	Law of Large Numbers; Central Limit Theorem	10.1-10.3	
15	12/7-12/9	Bonus Topics		
Finals	12/10-12/18	Finals	NA	Exam #3, TBD

Note

The instructor has the right to modify the syllabus if needed. If this occurs, students will be notified before the change occurs.

Acknowledgement: The content and structure of this course are modeled after materials from Drs. Robin Belton (Vassar College) and Ming-Wen An (Vassar College).